

# Scenario Based Answers

## Set Answers 2

### Scenario 1: Delivery Time Analysis

#### Q1. Calculate Q1 and Q3

##### Logic:

Q1 is 1<sup>st</sup> quartile of 25th percentile and Q3 is 3<sup>rd</sup> quartile of 75th percentile. They divide the data into four equal parts.

##### Formula:

$$Q1 = (n+1)/4$$
$$Q3 = 3(n+1)/4$$

#### Q2. Find IQR

##### Logic:

Interquartile Range (IQR) measures the spread of the middle 50% of the data.

##### Formula:

$$IQR = Q3 - Q1$$

#### Q3. Detect Outliers

##### Logic:

Outliers are unusual observations that lie far away from the majority of the data.

##### Formula:

$$\text{Lower Bound} = Q1 - 1.5(IQR)$$
$$\text{Upper Bound} = Q3 + 1.5(IQR)$$

If value < Lower Bound or If value > Upper Bound = Outlier.

## **Scenario 2: Student Score Analysis**

### **Mean**

#### **Logic:**

Mean is the average of all observations.

#### **Formula:**

$$\text{Mean} = \sum X / N$$

### **Median**

#### **Logic:**

Median is the middle value after arranging the data in ascending order.

#### **Formula:**

Odd:

$$\text{Median} = (n+1)/2$$

Even:

$$\text{Median} = \text{Middle1} + \text{Middle2} / 2$$

### **Mode**

#### **Logic:**

Mode is the value that occurs most frequently.

### **Why Median is Better?**

#### **Logic:**

Median is less affected by extreme values (outliers), whereas mean can be influenced significantly by very high or very low observations.

### **Scenario 3: Grocery Store Customer Analysis**

#### **Frequency Distribution**

##### **Logic:**

Frequency distribution shows how many times each value occurs in the dataset.

##### **Formula:**

Frequency = Count of Occurrences / Frequency

### **Scenario 4: Real Estate Model Analysis**

#### **Detect Multicollinearity**

##### **Logic:**

Multicollinearity occurs when independent variables are highly correlated with each other.

##### **Correlation Formula**

$$r = \text{Cov}(X, Y) / \sigma_X \sigma_Y$$

If:

$$|r| > 0.8$$

Multicollinearity may exist.

##### **VIF Formula**

$$\text{VIF} = 1 / (1 - R^2)$$

Interpretation:

VIF < 5 → Acceptable

VIF > 5 → Moderate

VIF > 10 → High Multicollinearity

## **Scenario 5: Medicine Effectiveness Study**

### **Hypothesis Testing**

#### **Logic:**

Hypothesis testing is used to determine whether the medicine has a significant effect compared to the placebo.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

#### **Test Used**

#### **Independent t-test**

Decision:

$$p < 0.05$$

Reject  $H_0$

$$p > 0.05$$

Accept  $H_0$

## **Scenario 6: Sales Outlier Detection**

### **IQR Method**

$$IQR = Q3 - Q1$$

$$\text{Lower} = Q1 - 1.5 * (IQR)$$

$$\text{Upper} = Q3 + 1.5 * (IQR)$$

### **Z-Score Method**

#### **Formula:**

$$Z = \frac{X - \mu}{\sigma}$$

Interpretation:

$$|Z| > 3 = \text{Outlier}$$

## **Scenario 7: Customer Satisfaction**

### **Mean**

$$\text{Mean} = \sum X / N$$

### **Median**

Median = Middle Value

### **Mode**

Mode = Most Frequent Rating Mode

### **Logic:**

Customer satisfaction can be summarized using mean, median, mode, and frequency distribution to understand the overall satisfaction level.